

A water-extract of the Korean traditional formulation Geiji–Bokryung–Hwan reduces atherosclerosis and hypercholesteremia in cholesterol-fed rabbits

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Abstract

Geiji–Bokryung–Hwan (GBH), a drug preparation consisting of five herbs of *Cinnamomi Ramulus* (Geiji), *Poria Cocos* (Bokryun), *Mountain Cortex Radicis* (Mokdanpi), *Paeoniae Radix* (Jakyak) and *Persicae Semen* (Doin), is a traditional Korean herbal medicine that is widely used in the treatment of atherosclerosis-related disorders. A water extract of GBH was found to scavenge 1,1-diphenyl-2-picrylhydrazyl (DPPH) radicals and inhibit low-density lipoprotein (LDL) oxidation more effectively than probucol, a well-known commercially available antioxidant. In order to evaluate the anti-atherogenic potential of this medication, New Zealand White (NZW) rabbits were fed a normal diet for 12 weeks, a high cholesterol diet, a high cholesterol diet containing 1% probucol or a high cholesterol diet containing 5% water-soluble extract of GBH. Both GBH and probucol reduced plasma cholesterol levels. LDLs from the GBH-treated group were more resistant to Cu²⁺-induced oxidation and contained more vitamin E than LDLs from the high cholesterol diet group. Endothelial damage, determined at week 6, was reduced by 55% in the GBH group ($P < 0.01$). GBH treatment reduced an atherosclerotic area in the abdominal aorta by 58% ($P < 0.05$) and cholesterol deposition in the thoracic aorta by 55% ($P < 0.05$). The severity of atherosclerosis in the GBH group was significantly reduced after an adjustment using cholesterol exposure as an index of the cholesterol-lowering effect. On the other hand, diet-induced hyperlipidemic rabbits were given water extract of GBH in doses of 50 (Group B) and 200 mg/kg (Group C) and compared with controls (Group A). At 40 days after intervention in groups A, B and C, total and LDL cholesterol levels were significantly lowered ($P < 0.01$). LDL/high density lipoprotein (HDL) ratio was also significantly decreased ($P < 0.01$). This study concludes that the reduction in atherosclerosis by GBH relies not

Abbreviations: GBH, Geiji–Bokryung–Hwan; LDL, low density lipoprotein; HDL, high density lipoprotein; OxLDL, oxidized LDL; DPPH, 1,1-diphenyl-2-picrylhydrazyl; BSA, bovine serum albumin; NZW, New Zealand white; TG, triglyceride; TBARS, TBA-reactive substance; EBA, Evans blue-albumin; BUN, blood urea nitrogen; AST, aspartate aminotransferase; ALT, alanine aminotransferase.

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only on its cholesterol-lowering effect but also more heavily on its antioxidant potential, which prevents endothelial damage and inhibits LDL oxidative modification in hypercholesterolemic animals.

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1. Introduction

The pathogenesis of atherosclerosis involves endothelium dysfunction, infiltration of monocytes, activation of monocytes into macrophages and smooth muscle cell proliferation [1]. Recent studies have demonstrated that oxidized low-density lipoprotein (OxLDL) plays an important role in the initiation and progression of atherosclerosis [2]. Minimally modified low-density lipoprotein (LDL) is capable of inducing gene expression in endothelial cells that may result in the acceleration of atherogenesis [3–5]. On further modification of the intima, OxLDL is taken up by macrophage scavenger receptors, gradually leading to the formation of foam cells and fibrous plaques [6]. Immunochemical studies have demonstrated that OxLDL is present in the atherosclerotic lesions of animals and humans [7,8].

Antioxidants that prevent LDL from oxidative damage may interrupt the progression of atherosclerosis. Vitamins E and C protect LDL from oxidative damage in vitro [9,10] and decrease the morbidity of coronary heart disease [11–14]. Probucol has been shown to inhibit atherosclerosis in non-human primates [15] and rabbits [16]. It has recently been demonstrated that *N,N'*-diphenyl-phenylene-diamine reduces atherosclerosis in cholesterol-fed rabbits and apoE-deficient mice [17,18]. The ability of lipophilic antioxidants to reduce atherosclerosis and the incidence of coronary heart disease is mainly due to protection of LDL from oxidative modification [14]. Since water-soluble antioxidants are unable to adhere to LDL particles, they are unable to provide this protection in the subendothelial space. However, data from several studies suggest that water-soluble antioxidants, such as vitamin C, are effective in inhibiting LDL oxidation via the preservation of endogenous antioxidants in LDL [19]. Recent studies have also shown that flavonoids, a class of water-soluble antioxidants, are useful in reducing the risk of coronary heart disease [20].

High levels of LDL and low levels of high-density lipoprotein (HDL) cholesterol are important coronary risk factors. Dietary control continues to be the anchor for the treatment of these conditions, but in many cases, it becomes necessary to add a drug [21]. Since none of the available agents fulfill the requirements of such a desired drug, there is a need to explore the possibility of inducing an effective, safe and inexpensive alternative [22].

Some traditional Korean prescriptions employed for a syndrome expressed in the oriental medical concept as chest paralysis and heartache are thought to be effective for angina pectoris. Therefore, we investigated the effects of an oriental medicinal prescription, Geiji–Bokryung–Hwan (GBH), that is comprised of five herbs of *Cinnamomi Ramulus* (Geiji), *Poria Cocos* (Bokryun), *Mountain Cortex Radicis* (Mokdanpi), *Paeoniae Radix* (Jakyak) and *Persicae Semen* (Doin) on anti-atherosclerosis activity. GBH, a well known herbal medicine used in the treatment of cardiovascular disorders (referred to as blood stasis in traditional Korean medicine), is widely used in China, Japan, Taiwan and Korea. Nonpolar extracts contain some antioxidative substances, which can inhibit platelet aggregation [23] and protect the myocardium against ischemia-induced derangement. An aqueous extract of GBH contains phenolic compounds that are effective in protecting liver microsomes, hepatocytes and erythrocytes against oxidative damage. For example, a potent hepatoprotective agent and water-soluble antioxidant are suggested [24].

It has previously been reported that the GBH acts as an antioxidant and antimutagen and to stimulate phase II drug-metabolizing enzymes (anti-initiation activity); it mediates anti-inflammatory effects and inhibits cyclooxygenase and hydroperoxidase functions (antipromotion activity); and it induces human promyelocytic leukemia cell differentiation (antiprogession activity). The inhibitory effect of GBH water extracts on the growth of cancer cell lines such as

HepG2 cell and Hep3B cell has been demonstrated. These data suggest that GBH extracts merit investigation as a potential cancer chemopreventive agent in humans, especially in hepatological cancers [25].

The purpose of the present study was to elucidate the potential of GBH in the inhibition LDL oxidation and the EtOH-fraction of GBH in the reduction of the severity of atherosclerosis in hypercholesterolemic rabbits. We also investigated the influence of doses of the water extract and ethanol extract in rabbits on cholesterol lipoproteins, atherosclerosis and tissue lipids.

2. Materials and methods

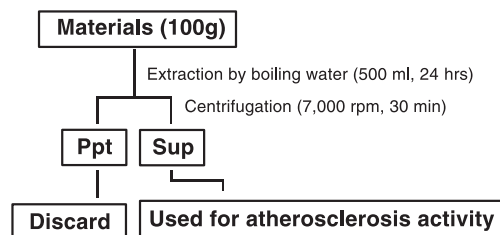
2.1. Materials

Probucol was obtained from ICN Biomedicals (Costa Mesa, CA) or from our supplies. Caffeic acid, 1,1,3,3-tetramethoxypropane, 1,1-diphenyl-2-picrylhydrazyl (DPPH), bovine serum albumin (BSA), tocopherol, retinyl acetate, Evans blue, Sudan IV and hematoxylin were purchased from Sigma.

GBH, a prescription formulation (No. G-27-5), was obtained from Dongguk University Oriental Medical Hospital, Kyungju, Korea (Scheme 1). In addition, for laboratory study, authentic plant materials were purchased from a local market and identified at the Oriental Medical Department, Dongguk University, Kyungju. Voucher specimens [OM-G1, OM-G2, OM-G3, OM-G4 and OM-G5] are on deposit at the Herbarium of Botany Department, Dongguk University, Kyungju, Korea.

Medicinal herb name	Dose amount (g)
<i>Cinnamomi Ramulus</i>	1.33
<i>Poria Cocos</i>	1.33
<i>Moutan Cortex Radicis</i>	1.33
<i>Paeoniae Radix</i>	1.33
<i>Persicae Semen</i>	1.33
Total	6.65

Scheme 1. Composition of GBH.



Scheme 2. Preparation of the water-extracts.

2.2. Animals and blood sample collection

Male New Zealand white (NZW) rabbits were obtained from the Animal Center of KCTC, Korea Research Institute of Bioscience and Biotechnology (KRIBB), Taejon, Korea. High fiber rabbit chow (as normal diet) was purchased from Purina Korea.

Adult albino rabbits (giant white strain) weighing 1.5–2.5 kg were procured from KCTC. Initially, they were given a controlled diet consisting of normal pellets for a period of acclimatization. Blood samples were collected from the marginal ear vein of each rabbit and the serum was separated for the estimation of lipids of total cholesterol, HDL cholesterol, LDL cholesterol and triglyceride (TG) according to general methods [22].

2.3. Preparation of water-soluble GBH

The dried plants of GBH were extracted with water (Scheme 2). Plant was cut into small sections, boiled twice in water for 30 min. The residues were removed by filtration and the combined filtrate was evaporated to obtain the desired concentration (2 g dry plant equivalent extract/ml). In a typical experiment, the plant was cut into pieces and macerated in distilled water. The liquid was decanted on day 4 and filtered. The filtrate was evaporated to dryness in an oven at 40 °C. The dried extract was then weighed (yield, 1.5%).

2.4. Ethanol extraction

A dried ethanolic extract of the GBH was also prepared. GBH was powdered and subjected to soxhlet extraction with 50% v/v ethanol in a Soxhlet apparatus. The extraction time was 16 h. The liquid extract collected in a flask was filtered. The ethanol was evaporated in a hot air oven at 40 °C to obtain a solid

residue. Finally, the ethanol extract was freeze-dried to give a powder. This powdered extract was administered to rabbits in the form of a watery suspension.

2.5. DPPH radical scavenging and inhibition of LDL oxidation

The DPPH radical-scavenging activities of the EtOH fraction, GBH and probucol were determined. In a typical run, 1 vol of acetate buffer (100 mmol/l, pH 5.5), 1 vol of ethanol and 0.5 vol of freshly prepared DPPH ethanolic solution (500 nmol/l) were mixed. After adding the test compound, the mixture was incubated at 25 °C for 90 min. The change in absorption at 517 nm was then determined.

Fasting plasma samples were collected from healthy male adults that had not been given vitamin supplements. LDL was obtained by ultracentrifugation from a density range of $1.019 < d < 1.063$ (adjusted by NaBr) [26]. LDL was extensively dialyzed against PBS (5 mmol/l phosphate buffer and 145 mmol/l NaCl, pH 7.4) at 4 °C under nitrogen for 24 h. In vitro LDL oxidation was carried out in a 96-well microtiter plate at 37 °C [27]. A probucol stock solution (2.0 mmol/l) was prepared in ethanol. It was diluted to 50 µmol/l into 10% aqueous ethanol before use as a positive control. The EtOH fraction and GBH were dissolved in PBS. A 50-µl aliquot of LDL (0.9 mg cholesterol/ml) in each well was preincubated with the test compound for 1 h. The final volume in each well was adjusted to 100 µl with PBS. LDL oxidation was initiated by adding CuSO₄ to a final concentration of 10 µmol/l. After incubation, 150 µl EDTA (2 mmol/l) was added. A 100-µl portion of the mixture was then transferred to a minivial containing 0.9 ml of 2-propanol. The precipitates were removed by centrifugation. The concentration of conjugated dienes in the supernatant was determined by absorption at 234 nm.

2.6. Animal treatment

Male NZW rabbits were randomly divided into four groups. Five rabbits in the control group were fed a normal diet. Seven rabbits in the high cholesterol group were fed a high cholesterol diet (normal diet supplemented with 1% cholesterol and 4% corn oil). Seven rabbits in the probucol group were fed a high cholesterol diet plus 1% (wt/wt) probucol [28]. Seven

rabbits in the GBH group were fed a high cholesterol diet plus 50 or 200 mg/kg GBH. The selection of GBH dose was based on its antioxidant activity compared to that of probucol in Cu²⁺-induced LDL oxidation (Table 1). Diets and drinking water were provided ad libitum. Animals were bled periodically for measurement of plasma cholesterol and liver function. During the 12-week feeding period, we adhered to the guidelines for care and use of laboratory animals.

2.7. Plasma cholesterol and lipoprotein analysis

Fasting plasma levels were obtained from animals after 0, 1, 3, 6, 9 and 12 weeks of feeding. Concentrations of plasma and lipoprotein cholesterol were determined enzymatically [29]. Cholesterol exposure was calculated as the area under the curve of plasma cholesterol vs. time. For the lipoprotein analysis, an aliquot of plasma was prestained with Sudan black for 20 min. Electrophoresis was performed using a polyacrylamide gel with a discontinuous gradient (2% in the upper layer and 3% in the lower layer).

2.8. Susceptibility of LDL oxidation ex vivo

Fasting blood samples were collected into evacuated tubes (K3-EDTA) from rabbits after 12 weeks of feeding. The LDL fraction ($1.019 < d < 1.063$) was collected by ultracentrifugation and extensively dialyzed in PBS (pH 7.4). The concentration of dialyzed LDL was adjusted to 0.9 mg of cholesterol/ml. It was incubated with CuSO₄ (final concentration, 25 µmol/l) at 37 °C. A subsequent assay was performed according to the procedure of Puhl et al. [30] with minor modifications. The formation of malonaldehyde was determined as TBA-reactive substance (TBARS) [27]. In brief, an aliquot of the incubation solution was added to a TBA solution (0.6%, wt/vol), followed by heating at 95 °C for 45 min. After centrifugation, TBARS in the supernatant was measured by absorption at 532 nm.

2.9. Vitamin E content in LDL

Without dialysis, a 200-µl aliquot of LDL was added to an equal volume of ethanol containing 7.5 µmol/l retinyl acetate as an internal standard. The

Table 1
Effect of GBH on lipid profile

Lipoprotein lipid	Initial	Atherogenic diet	Treatment with or without GBH	
			20 days	40 days
<i>Control group A (n = 7)</i>				
Total cholesterol	154.6 ± 14	875 ± 121	623 ± 46	564 ± 73
LDL cholesterol	83 ± 6	843 ± 112	582 ± 54	512 ± 54
HDL cholesterol	43 ± 7	86 ± 7	79 ± 5	63 ± 5
Triglycerides	98 ± 21	127 ± 13	135 ± 21	112 ± 12
Total/HDL cholesterol	4.4 ± 0.3	10.3 ± 0.6	9.5 ± 0.5	10.3 ± 0.6
LDL/HDL cholesterol	2.5 ± 0.4	10.1 ± 0.8	8.4 ± 0.6	8.6 ± 0.8
<i>GBH (50 mg/kg) group B (n = 7)</i>				
Total cholesterol	160 ± 15	980 ± 95	587 ± 65	354 ± 24**
LDL cholesterol	110 ± 32	965 ± 74	532 ± 32	283 ± 27**
HDL cholesterol	30 ± 3	94 ± 6	51 ± 4	38 ± 4
Triglycerides	87 ± 5	157 ± 23	80 ± 9	72 ± 43
Total/HDL cholesterol	6.0 ± 0.4	12.6 ± 1.6	13.8 ± 2.1	9.3 ± 1.2
LDL/HDL cholesterol	4.1 ± 0.3	9.5 ± 0.6	11.6 ± 1.6	7.5 ± 0.6
<i>GBH (200 mg/kg) group C (n = 7)</i>				
Total cholesterol	174 ± 21	921 ± 102	402 ± 43	265 ± 32***
LDL cholesterol	121 ± 11	734 ± 76	295 ± 34	176 ± 15***
HDL cholesterol	45 ± 4	78 ± 9	44 ± 6	36 ± 4
Triglycerides	87 ± 8	137 ± 15	121 ± 21	106 ± 17
Total/HDL cholesterol	4.8 ± 0.5	10.8 ± 1.6	9.4 ± 0.6	5.8 ± 0.7*
LDL/HDL cholesterol	3.1 ± 0.3	9.5 ± 0.6	7.5 ± 1.2	4.6 ± 0.4**

All values are in mg/100 ml ± S.E.M. Comparison of control group with treatment groups B and C (*t*-test).

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

mixture was extracted with *n*-hexane containing BHT (0.4 mg/ml) in the dark. The hexane layer (1.5 ml) was dried under a stream of nitrogen, and the residues were resuspended in 100 µl of mobile phase (acetonitrile/tetrahydrofuran, 70:30, vol/vol). After filtration through glass wool, the content of vitamin E (as α -tocopherol) was determined by reversed-phase HPLC [31].

2.10. Endothelial damage

Endothelial damage was determined in separate animals after 6 weeks of an identical feeding schedule. Quantification was based on the leakage of Evans blue-albumin (EBA) conjugate as a macromolecular tracer [32]. Animals were anesthetized with a mixture of ketamine and xylazine hydrochloride solution. The right femoral artery, vein and right carotid artery were cannulated. Fifteen milliliters of

Evans blue-albumin (EBA) solution (200 mg BSA/ml) was slowly injected into the femoral vein. Exactly 5 min after injection, an overdose of pentobarbital was given. Shortly before euthanasia of the animal, heparin (5000 USP units) was injected intravenously through the femoral vein to prevent intravascular blood coagulation. The arterial system was perfused immediately with heparinized saline (2 U/ml) via the right carotid artery at a pressure of 120 mm Hg. After the fluid was drained from the femoral artery and the vein turned clear, the perfusate was switched to a neutral-buffered 10% formalin solution. Perfusion was continued at the same pressure for another 10 min as a preliminary perfusion fixation. For further fixation, the thoracic aorta was excised and immersed in 10% formalin for 1 h. The adventitial tissue and branches were carefully removed. Each aortic specimen was cut longitudinally and dissected into six pieces (100 mm²/segment). The

specimens were stained with Harris' hematoxylin for 1 min. An epifluorescence microscope was used for detection and quantification of EBA leaky foci. The EBA fluorescence was examined using a green excitation at 546 nm.

2.11. Extent of atherosclerosis and cholesterol deposition

After 12 weeks of feeding, the animals were killed, and the aortas, from the aortic arch to the iliac bifurcation, were collected and any adhering tissue removed. Abdominal aortas were stained with a solution of Sudan IV to visualize the lesion area [17]. Sudanophilic areas were photographed and measured by computer-assisted planimetry. Thoracic aortas were weighed, minced and extracted with a mixture of chloroform and methanol (2:1, vol/vol) for total lipids. Aliquots of the lipid extracts were saponified in an ethanolic KOH solution. Cholesterol was extracted into *n*-hexane, and the levels determined by means of a microscale method described by Abell et al. [33].

2.12. Oral administration of GBH

Eighteen rabbits were made hyperlipidemic by feeding an atherogenic diet containing 60 g rizka powder, 50 g wheat flour, 5 g hydrogenated ground nut oil, 4 g salt and 400 mg/kg cholesterol powder for 20 days. They were then divided into three groups of six animals each. One group was given a normal diet (Group A), the second and third groups were given a normal diet along with GBH extract 50 (Group B) and 200 mg/kg (group C), respectively.

2.13. Statistical analysis

Results are reported as the mean $\pm$ S.E.M. The differences in number of EBA leaky foci between groups were analyzed by the paired *t*-test. Other statistical analyses were performed using the unpaired *t*-test or ANOVA. A value of $P < 0.05$ was considered to be statistically significant.

The values for various lipids are given as the mean $\pm$ S.E.M. Differences between the experimental and control group were determined by the unpaired *t*-test and $P < 0.05$ was considered to be significant.

3. Results

3.1. Effect of GBH on total cholesterol

Total cholesterol after feeding the atherogenic diet increased significantly from 154.6 ± 14 to 875 ± 121 mg/ml (Table 1). On substituting the atherogenic diet, the A normal diet for 40 days, total cholesterol decreased to 564 ± 73 mg/ml in control Group A.

The GBH extract administered in doses of 50 and 200 mg/kg with a normal diet to hypercholesterolemic rabbits showed a reduction in total cholesterol at 40 days to 354 ± 24 mg/ml for Group B ($P < 0.01$) and 265 ± 32 mg/ml in Group C ($P < 0.001$) (Table 1).

3.2. Effect of GBH on LDL cholesterol

LDL cholesterol increased from a base line value of 83 ± 6 to 843 ± 112 mg/ml after feeding the atherogenic diet (Table 1). On substituting with a normal diet, LDL cholesterol levels decreased to 512 ± 54 mg/ml at 40 days for Group A. GBH extract reduced LDL cholesterol to 283 ± 27 mg/ml in Group B ($P < 0.01$) and 176 ± 15 mg/ml in Group C ($P < 0.001$).

3.3. Effect of GBH on HDL cholesterol

HDL cholesterol increased from a base line value of 43 ± 7 to 86 ± 7 mg/ml in atherogenic diet-fed rabbits. Substitution with a normal diet resulted in a reduction to 63 ± 5 mg/ml at 40 days. A GBH extract in doses of 50 and 200 mg/kg decreased HDL cholesterol levels to 38 ± 4 mg/ml in Group B and 36 ± 4 mg/ml in Group C (Table 1).

3.4. Effect of GBH on total cholesterol: HDL ratio

The total cholesterol/HDL ratio increased from 4.4 ± 0.3 to 10.3 ± 0.6 after feeding the atherogenic diet. Substitution with a normal diet reduced this ratio to 8.6 ± 0.8 in group B. GBH reduced the ratio to 9.3 ± 1.2 and 5.8 ± 0.7 in Group C ($P < 0.05$) (Table 1).

3.5. Effect of GBH on LDL/HDL ratio

LDL/HDL ratio increased from a base line value of 2.5 ± 0.4 to 10.1 ± 0.8 after feeding atherogenic diet.

Table 2

IC₅₀ values and relative potency of probucol, EtOH fraction and GBH

	DPPH scavenging		Inhibition of LDL oxidation	
	IC ₅₀ (μg/ml)	Relative potency	IC ₅₀ (μg/ml)	Relative potency
Probucol	55	1.0	2.65	1.0
EtOH fraction	13	4.1	0.91	3.2
GBH	221	0.24	16.3	0.17

After replacement with a normal diet, the ratio was reduced to 8.6 ± 0.8 after 40 days in control. GBH in Group B did not significantly decrease this ratio (7.5 ± 0.6) while in Group C, a significant decrease to 4.6 ± 0.4 occurred.

The results in Table 1 also show that there was no difference in TG levels at 40 days intervention in Groups B and C as compared to the controls.

3.6. DPPH scavenging and inhibition of LDL oxidation by GBH

Both the EtOH fraction and GBH exhibited free radical-scavenging activity, as evidenced by the DPPH assay, and were effective in preventing Cu²⁺-

induced LDL oxidation (Table 2). On the basis of IC₅₀ values, GBH was 4.1 times more potent than probucol in the DPPH assay and 3.0 times more potent than probucol in the Cu²⁺-induced LDL oxidation assay. It was estimated that the EtOH fraction accounted for up to 72% of the antioxidant activity in GBH since the EtOH fraction was 16 times more potent than GBH in both assay systems.

No detectable change was found in wavelength (max) or molecular absorption (log) of the EtOH fraction at 50 μg/ml when it was coincubated with LDL (100 μg/ml) in PBS: max (log) = 294 (4.24), 316 (4.05) and 341 (4.26). No indication of the direct binding of EtOH fraction with LDL particles in solution was found. Similar observations were also found when fresh, extensively dialyzed human LDL preparations were used.

Table 3

Effect of GBH on body weight and AST, and plasma cholesterol, LDL cholesterol and vitamin E content in LDL of rabbits treated for 15 weeks

	Normal (n = 5)	High cholesterol (n = 6)	Probucol (n = 5)	GBH (n = 6)
Body weight (kg)	2.7 ± 0.4	3.2 ± 0.4	2.9 ± 0.5	3.0 ± 0.4
AST (U/L)	23 ± 2.6	38 ± 5.1	58 ± 8.4	22 ± 6.3
Plasma cholesterol	1.3 ± 0.3	70.2 ± 5.7*	35.7 ± 4.5	60.3 ± 6.4
LDL cholesterol	0.43 ± 0.12	12.5 ± 2.1	13.2 ± 2.4	10.4 ± 3.2
Vitamin E	15.2 ± 2.2	10.5 ± 2.5**	7.8 ± 0.13**	23.3 ± 2.2

Plasma cholesterol and LDL cholesterol were expressed as millimolar values. The vitamin E content in LDL was expressed as nanomoles α-tocopherol per micromole LDL cholesterol. The value was 14.2 ± 2.4 nmol α-tocopherol/μmol LDL cholesterol before the animals were divided for the feeding experiment. Data represent mean ± S.E.M. of an individual group.

* $P < 0.05$ vs. corresponding probucol group.

** $P < 0.01$ vs. corresponding GBH group.

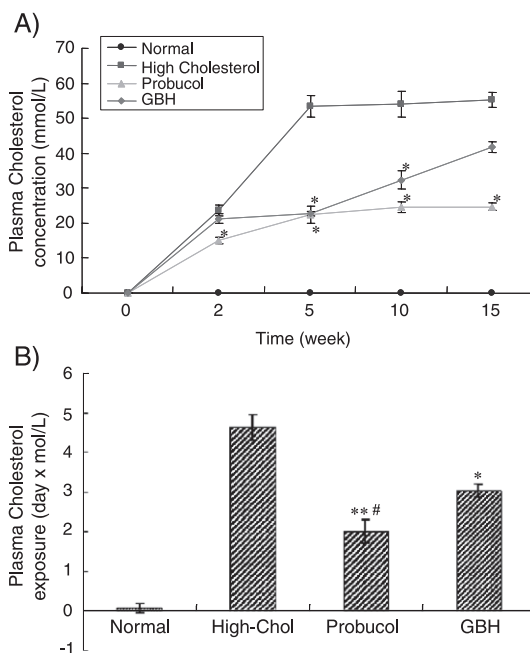


Fig. 1. Plasma cholesterol (Chol) concentration (A) and Chol exposure (B). The groups are as follows: normal ($n = 5$, ●), high Chol ($n = 6$, ■), probucol ($n = 6$, ▲) and GBH ($n = 6$, ◆). Plasma Chol concentrations were lower in the GBH- and probucol-treated groups compared with the high Chol group (two-way ANOVA, $P < 0.01$). * $P < 0.05$ vs. high Chol in panel A. Chol exposure was defined as the area under the curve of plasma Chol vs. time. * $P < 0.05$ and ** $P < 0.01$ vs. high Chol; # $P < 0.05$ vs. GBH in panel B.

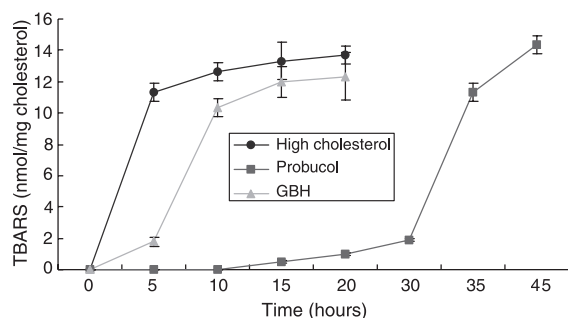


Fig. 2. Time course of Cu^{2+} -induced LDL oxidation. TBARS values were measured by absorption at 532 nm and are expressed as nanomoles of malonaldehyde per milligram LDL cholesterol (nmol/mg Chol). Groups are as follows: high cholesterol ($n=5$), probucol ($n=5$) and GBH ($n=6$). Each data point represents the mean $\pm$ S.E.M. for an individual group.

3.7. Results on animals

No differences in body weights were detected at the end of the 12-week feeding period (normal, 2.7 ± 0.4 kg; high cholesterol, 3.2 ± 0.4 kg; probucol, 2.9 ± 0.5 kg; and GBH, 3.0 ± 0.4 kg). Hypertriglyceridemia and hyperglycemia occurred in all groups of animals fed with the high cholesterol diet; neither probucol nor GBH significantly prevented the elevation of plasma triglyceride and glucose. Probucol treatment caused an elevation in aspartate amino-

transferase (AST) values (normal, 23 ± 2 U/l; high cholesterol, 38 ± 5 U/l; and probucol, 57 ± 8 U/l). The GBH group had an AST value (22 ± 6 U/l) lower than that of the high cholesterol group ($P < 0.05$) (Table 3).

There were no significant differences in alanine aminotransferase (ALT), electrolyte, blood urea nitrogen (BUN) or creatinine values among the three cholesterol-fed groups. GBH treatment did not cause any additional abnormality, as indicated by the biochemical parameters employed in this study.

3.8. Plasma cholesterol and aortic cholesterol exposure

Plasma cholesterol levels were increased during the feeding period of rabbits that had been fed a high cholesterol diet. GBH and probucol treatment attenuated the increase in plasma cholesterol (Fig. 1). Cholesterol exposure, defined as the area under the curve of plasma cholesterol vs. time, was determined in order to estimate the effect of cholesterol lowering on atherosclerosis reduction. GBH caused a 32% decrease and probucol caused a 41% decrease in cholesterol exposure compared with the high cholesterol group. The reduction in plasma cholesterol by GBH and probucol occurred predominantly in β -VLDL ($d < 1.019$) (data not shown). Probucol treat-

Table 4
Endothelial damage determined by leakage of EBA

Segment	Endothelial damage (leakage foci/mm ²)			
	Normal	High cholesterol	Probucol	GBH
S1	0.70	2.43 (1.32)	0.87 (0.04)	1.32 (0.25)
S2	0.73	2.54 (1.43)	1.54 (0.76)	2.12 (1.23)
S3	0.80	3.23 (2.36)	2.04 (1.03)	2.16 (1.32)
S4	0.83	3.54 (2.56)	2.54 (1.67)	2.21 (1.54)
S5	0.88	4.23 (4.52)	2.45 (1.43)	2.54 (1.87)
S6	0.85	6.54 (6.32)	4.76 (3.31)	3.21 (2.21)
Mean $\pm$ S.E.M.	0.80.3 $\pm$ 0.05	3.72 $\pm$ 0.41	2.34 $\pm$ 0.54	2.10 $\pm$ 0.26
Reduction	—	—	— 25%*	— 36%*
Mean $\pm$ S.E.M. (corrected)	—	2.98 $\pm$ 0.54	1.65 $\pm$ 0.23	1.32 $\pm$ 0.21
Reduction	—	—	— 47%*	— 46%*
P value	—	—	$P < 0.05^*$	$P < 0.01^*$

Endothelial damage was determined after 5 weeks feeding. S1–S6 denote the segment (100 mm²/segment) of thoracic aorta from the distal end (diaphragm) to the proximal end (ligamentum arteriosum). Values in parentheses indicate the corrected numbers of leaky foci obtained after subtracting the corresponding values of the normal group from the treated group. Paired *t*-tests were used for statistical analysis.

*Vs high cholesterol group.

ment decreased HDL cholesterol levels at weeks 5 and 10 compared with the levels in the high cholesterol group ($P<0.01$) and levels before treatment (week 0 vs. week 10, $P<0.01$). Throughout the feeding, no difference in HDL cholesterol was observed between the GBH and high cholesterol groups.

3.9. Oxidative susceptibility in LDL

LDLs collected from GBH- and probucol-treated groups were more resistant to oxidation than those from the high cholesterol group (Fig. 2). The time required to reach half-maximum TBARS production was defined as antioxidant activity. The antioxidant activity of probucol (34.7 ± 3.6 h) was the highest among the three cholesterol-fed groups. The antioxidant activity of the GBH-treated group (6.0 ± 0.4 h) was significantly higher than that of the high cholesterol group (1.5 ± 0.1 h).

3.10. Vitamin E content in LDL

Cholesterol feeding decreased the vitamin E content of LDLs. In the high cholesterol group, the value (9.4 ± 1.4 nmol/ μ mol cholesterol) was 34% less than that in the normal group (16.4 ± 1.5 nmol/ μ mol cholesterol) ($P<0.05$) (Table 3). Probucol treatment further decreased vitamin E content in LDLs by 17%. The vitamin E content of LDLs collected from the GBH-treated group was 21.3 ± 1.4 nmol/ μ mol cholesterol, a 2.2-fold higher value than that in the high cholesterol group ($P<0.01$).

3.11. Protection of endothelial cellular damage

Endothelial damage, as determined by the number of EBA leaky foci in the thoracic aortas, was increased by cholesterol feeding. The number of leaky foci was significantly increased in the high cholesterol group (4.51 ± 0.35 foci/ mm^2) compared with the normal group (0.95 ± 0.10 foci/ mm^2) ($P<0.05$) (Table 4). Compared with the high cholesterol group, endothelial damage was reduced by 44% in the GBH group ($P<0.01$) and by 37% in the probucol-treated group ($P<0.05$). To reveal the protective effect of GBH against endothelial damage caused by cholesterol feeding, the number of leaky foci was corrected by subtracting the basal value of the normal group

from the value of the treatment group value. GBH treatment decreased cholesterol feeding-induced endothelial damage by 49% ($P<0.01$).

3.12. Reduction of atherosclerotic area and cholesterol deposition

Severe atherosclerosis developed in animals after 15 weeks of a high cholesterol diet. The atherosclerotic area in the abdominal aorta was $36.2 \pm 1.8\%$ and cholesterol deposition in the thoracic aorta was 26.3 ± 3.1 mg/g in the high cholesterol group (Fig. 3). Probucol treatment resulted in a 70% decrease in atherosclerotic area and a 46% decrease in cholesterol deposition. GBH treatment caused a 48% decrease in the atherosclerotic area and a 43% decrease in cholesterol deposition. Both decreases were statistically significant ($P<0.01$) compared with the high cholesterol group values.

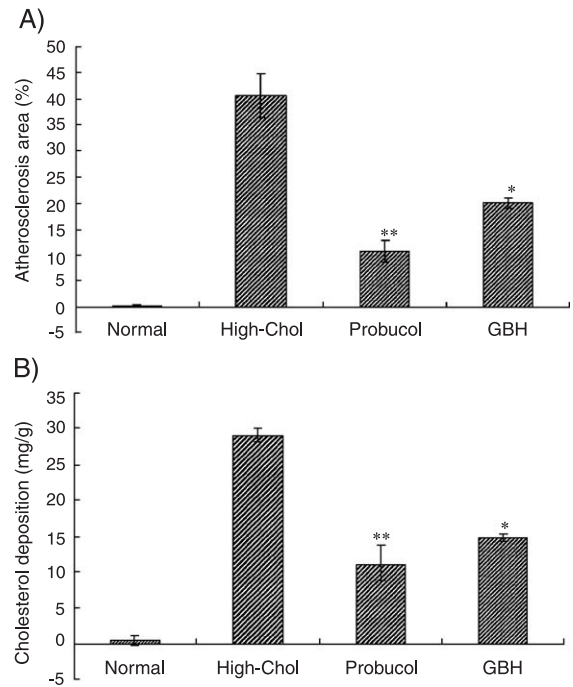


Fig. 3. Atherosclerotic area percentage in abdominal aorta (A) and cholesterol (Chol) deposition in thoracic aorta (B). ** $P<0.01$ vs. high Chol group ($n=5$) and * $P<0.05$ for probucol ($n=5$) vs. GBH ($n=6$) in panel A. * $P<0.05$ and ** $P<0.01$ vs. high Chol group ($n=5$) in panel B.

4. Discussion

This study clearly demonstrates that the GBH in doses of 50 and 200 mg/kg significantly reduced total and LDL cholesterol levels in hypercholesterolemic rabbits. At 200 mg/kg, it also reduced total/HDL and LDL/HDL ratios. The findings herein show that while the dosage forms were effective in decreasing total and LDL cholesterol, the total/HDL and LDL/HDL ratios were influenced only by a higher dose of 200 mg/kg. This finding may have important therapeutic implications. GBH decreased the fat content in both the liver and heart. This indicates a decreased availability of fats to these organs. This could be due to the decreased absorption of fats from the intestines due to the presence of fiber and some other effective substances in GBH. The resulting lowering of total and LDL cholesterol can thus be explained [34].

The present study showed that water-soluble GBH, which contains phenolic antioxidative substances as a potent antioxidants, scavenged DPPH radicals and inhibited Cu^{2+} -induced LDL oxidation. GBH feeding reduced endothelial damage and the severity of atherosclerosis in cholesterol-fed NZW rabbits. LDLs from GBH-treated animals contained more vitamin E and were more resistant to oxidation *ex vivo*.

The content of GBH in the high cholesterol diet was 4.5%. Since GBH contains only 14.5% antioxidative substances, the actual concentration of antioxidative substance in the chow was only 0.68%, a value much lower than that of probucol (1%). The DPPH-scavenging activity of the water-soluble antioxidant was dependent on its oxidizable functional group. GBH exerted radical-scavenging activities effectively in the DPPH system. However, the antioxidant activity tested in the LDL system was further dependent on interaction with LDL particles [35]. It is interesting to note that the relative potency of an antioxidative substance to GBH remained unchanged in both assay systems. This provides indirect evidence that indicates that the origin of the antioxidant activity of GBH was, in fact, predominantly an antioxidative substance. Being water-soluble and negatively charged, GBH is less likely to bind to LDL through the Cu^{2+} -binding sites. At a 36-fold higher concentration than its IC_{50} value, the phenolic

antioxidative substance (EtOH fraction) did not exhibit a detectable wavelength shift or absorption change when co-incubated with LDL from the high cholesterol group.

The severity of the atherosclerotic area in the abdominal aorta predicts the severity of cholesterol deposition in the thoracic aorta, and vice versa. A linear correlation between atherosclerotic area and cholesterol deposition was found. Cholesterol exposure is a known risk factor of atherosclerosis in cholesterol-fed rabbits [17]. To demonstrate the influence of factors other than an elevation in plasma cholesterol, the atherosclerotic area was normalized by cholesterol exposure. Results showed that GBH significantly reduced the atherosclerotic area per unit of cholesterol exposure. Atherosclerotic areas under similar amounts of cholesterol exposure were less extensive in the GBH-treated (–45%) group than in the high cholesterol group. We therefore conclude that the anti-atherosclerotic effects of GBH rely not only on cholesterol-lowering effects but also on their antioxidant activities as well.

Endothelial dysfunction is an early event in the pathogenesis of atherogenesis [4]. The present study demonstrates that GBH treatment significantly reduces endothelial damage. The role of phenolic antioxidative substances in resisting LDL oxidation in the subendothelial space could be the result of the preservation of vitamin E in LDL particles in the circulation [36]. Since the vitamin E content of LDL was higher in the GBH-treated group than in the high cholesterol group and since the GBH fraction did not contain vitamin E or C, we conclude that antioxidative substances in the GBH may indirectly protect LDL from oxidative modification by scavenging free radicals in the bloodstream of cholesterol-fed rabbits, thus sparing the vitamin E in LDL from being oxidized.

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